DermaPod© Subsystem RFQ Analysis

1. Optical & Geometric Summary

- **Resolution Target:** 30 px/mm baseline resolving power on skin surface, scalable up to 60 px/mm.
- **Depth of Field (DoF):** Maintained sharp focus across a body curvature of ≥ 100 mm.
- **Circle of Confusion (CoC):** Baseline CoC of 25 µm taken/assumed (not specified in RFQ).

Reference Parameters Used in Calculations:

- **Lens Aperture:** f/8
- **Body Curvature Target:** 100 mm
- **Mechanical Settle Time:** 2.0 seconds per step
- **Turntable & Column Rotation Time:** 5.0 seconds per transit
- **Focus Slice Exposure Time:** 0.2 seconds

2. 3D Simulation & Architectural Evaluation

Comparative review of the simulated gantry models using direct 3D visualizer captures:

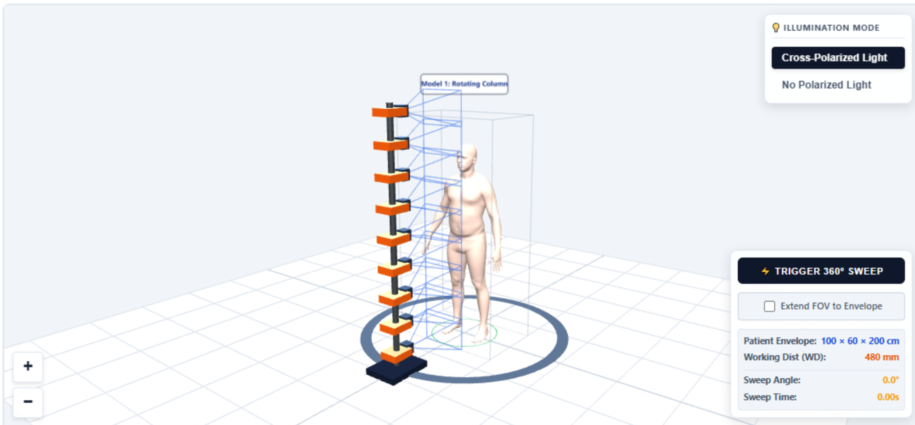
Model 1: Option A — Single-Column Rotating Gantry

- **Mechanical:** Single vertical column rotating 360° on an elliptical track.
- **Acquisition:** Performs a continuous 360° rotational sweep around the patient to capture overlapping frames and reconstruct a full 3D model.
- **Cost:** Minimal camera count, lowest BOM cost.
- **Risk:** Drive vibration during continuous rotation; patient postural sway/breathing during rotation.
- **Timing Considerations:** Model 1 sweep timing and rotational speed should be explored and optimized based on Structure-from-Motion (SfM) algorithms to ensure proper overlap.
- **Depth Assistance:** Auxiliary sensors like solid-state LiDAR may need to be integrated to provide absolute reference depth maps and correct for rotational mapping drift.
- **Artifact Risks:** The rotating column movement can trigger tracking reactions or balance adjustments, which needs to be evaluated for patient motion artifacts.

Model 1 Sensor Optimization Matrix (Target: 30 px/mm, CoC = 25 µm, Aperture = f/8, Depth Envelope = 0.6m):

Sensor Profile	FOV (W×H)	Total Array FOV	Avg Env Density	DoF (CoC=25µm)	Cams	Total Cost
GMAX3265 (65MP)	327mm × 216mm	327mm × 2052mm	21.3 px/mm	82.8 mm	11	$71,500
100MP Industrial	385mm × 289mm	385mm × 2002mm	24.5 px/mm	34.1 mm	8	$64,000

Sony IMX661 (127MP)	445mm × 317mm	445mm × 2190mm	25.4 px/mm	41.1 mm	8	$96,000

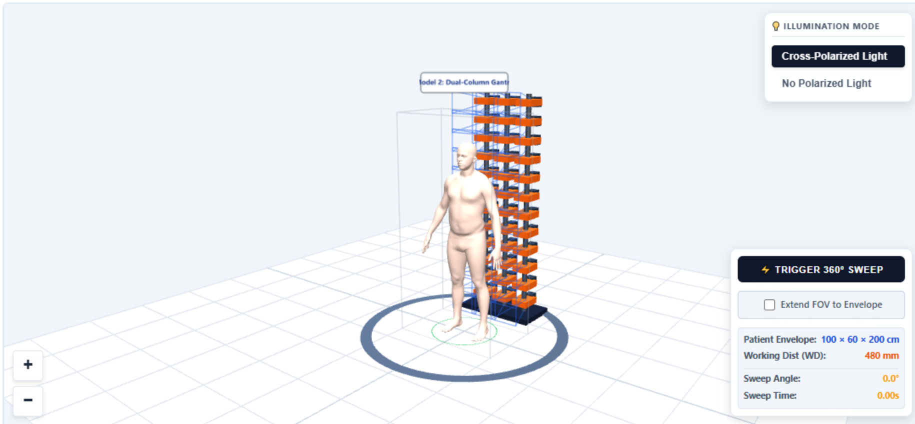


Model 2: Option B — Dual-Column Parallel Gantry

- **Mechanical:** Multiple parallel columns (upgraded to 3 columns to span the horizontal 1000mm width at 30 px/mm resolving power) on an elliptical track.
- **Acquisition:** Performs a segmented step-by-step horizontal rotation sweep, stopping at 4 discrete poses (0°, 90°, 180°, 270°) to capture complete patient coverage.
- **Cost:** High camera count and high BOM cost.
- **Patient Pose Guidelines:** Patient must dynamically adjust leg stance (forward/backward) and stretch hands based on the active camera pose location to prevent occlusions.
- **Risk:** Calibration complexity for multi-column geometries; high data ingestion bandwidth.
- **Timing Considerations:** Scan time is dictated by the column rotation transit and stop exposure time. Baselines assume 5.0s rotation transit time and 2.0s settle time per step.

Model 2 Sensor Optimization Matrix (Target: 30 px/mm, 3 Columns, CoC = 25 µm, Aperture = f/8):

Sensor Profile	FOV (W×H)	Total Array FOV	Avg Env Density	DoF (CoC=25µm)	Cams Needed	Total Cost	Scan Time
GMAX3265 (65MP)	327mm × 216mm	883mm × 2052mm	21.3 px/mm	82.8 mm	33 (3 × 11)	$214,500	26.2s
100MP Industrial	384mm × 288mm	1037mm × 2002mm	24.5 px/mm	34.0 mm	24 (3 × 8)	$192,000	27.8s
Sony IMX661 (127MP)	442mm × 315mm	1193mm × 2190mm	25.4 px/mm	40.6 mm	24 (3 × 8)	$288,000	27.8s

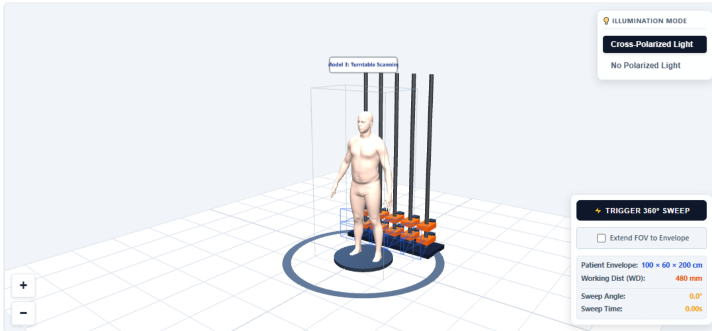


Model 3: Option C — Turntable Scanning Gantry

- **Mechanical:** Static columns with motorized vertical carriages; patient stands on a rotating turntable.
- **Acquisition:** Turntable rotates patient to 4 poses (0°, 90°, 180°, 270°); carriages sweep vertically at each stop.
- **Cost:** Low camera count, low BOM cost.
- **Patient Pose Guidelines:** Patient must dynamically adjust leg stance (forward/backward) and stretch hands based on the active camera pose location to prevent occlusions.
- **Timing Considerations:** Scan time is dependent on turntable rotation speed and vertical carriage travel steps. Baselines assume 5.0s turntable rotation time and 2.0s settle time per vertical sweep step.

Model 3 Sensor Optimization Matrix (Target: 30 px/mm, CoC = 25 µm, Aperture = f/8, Carriage Travel):

Sensor Profile	FOV (W×H)	Total Array FOV	Motor Steps	Avg Env Density	DoF (CoC=25µm)	Cams Needed	Total Cost	Scan Time
GMAX3265 (65MP)	326mm × 216mm	1110mm × 550mm	9 steps	21.3 px/mm	82.8 mm	12 (4 × 3)	$78,000	115.8s
100MP Industrial	394mm × 280mm	1004mm × 715mm	7 steps	24.5 px/mm	32.7 mm	9 (3 × 3)	$72,000	104.6s
Sony IMX661 (127MP)	444mm × 316mm	1133mm × 807mm	6 steps	25.4 px/mm	41.1 mm	9 (3 × 3)	$108,000	91.8s


odel 3: Turntable Scannin
ILLUMINATION MODE
Cross-Polarized Light
No Polarized Light
TRIGGER 360° SWEEP
Extend FOV to Envelope
Patient Envelope: 100 × 60 × 200 cm
Working Dist (WD): 480 mm
Sweep Angle: 0.0°
Sweep Time: 0.00s
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